A HOMEOMORPHISM ONTO A CLOSED SUBSET PRESERVES BEING NOWHERE DENSE

THE STACKS PROJECT AUTHORS

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Lemma 0.1. *Let $f : X \to Y$ be a continuous map of topological spaces. Let $T \subset X$ be a subset. If f is a homeomorphism of X onto a closed subset of Y and T is nowhere dense in X, then also $f(T)$ is nowhere dense in Y.*

Proof. Because $f(X)$ is closed in Y and f is a homeomorphism of X onto $f(X)$, we see that the closure of $f(T)$ in Y equals $f(\overline{T})$. Hence if $V \subset Y$ is open and contained in the closure of $f(T)$, then $U = f^{-1}(V)$ is open and contained in $\overline{T}$. Hence $U = \emptyset$, which in turn shows that $V = \emptyset$ as desired. □

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